WATER RESOURCES, WELLS & PUMPS · NOTE WR08

Economics of Irrigation — Sanction Criteria, Benefit–Cost Ratio and Water Rates

Course topics: IDE-514 T2, IDE-514 T3, IDE-514 T7

Economic justification

- Irrigation projects need large investments, and casual analyses and unjustified benefit claims are common. A project is economically justified only if it shows a profit after counting direct and indirect benefits and costs, with returns clearly above the cost of all inputs.
- Treat water projects as investments that create productive assets, with benefits over a long service life and well-defined sequences of investment and production; count costs and benefits over the whole service period.
- **Coverage:** surface and groundwater irrigation, water supply, drainage, land reclamation, salinity prevention and flood control.
- **Ownership:** irrigation is a State subject under the Constitution. States plan, execute, own and operate the projects — except open wells, shallow tubewells and some deep tubewells owned by individual cultivators.

Criteria for sanctioning projects

1. **From 1866:** loans were sanctioned for works expected to be remunerative, on cost–benefit tests.
2. **1955–61:** Planning Commission studies showed large benefits — double cropping, diversification, better crops, higher yields, incomes and employment — plus indirect benefits: processing and consumer industries, retail trade, transport and communications. Total benefits often exceeded the Government's direct revenue from water charges.
3. **A committee recommended the benefit–cost ratio** instead of the financial-return test. For simplicity it left out indirect benefits. Benefit = increase in annual value of farm production minus cultivation cost (before vs after irrigation). Cost = annual interest on capital + depreciation + operation and maintenance.
4. **1964 Committee:** recommended the economic-benefit criterion; the Government accepted it.
5. **Irrigation Commission (1972):** endorsed the benefit–cost ratio as superior, because financial returns depend on water charges that can be changed arbitrarily, while the B/C ratio reflects true economic worth.

Minor schemes — open wells, shallow and deep tubewells, small tanks, small canals in hills and pumping from rivers and drains — are usually taken up by individuals or cooperatives on loans; schemes needing large investment or complex engineering stay in the public sector.

Irrigation charges

Component	Basis
Water rate	Kind and extent of the crop
Increment in land revenue	Increased annual benefit from irrigation
Betterment levy	Government's share of the rise in land value
Irrigation cess	Annual charge per hectare of irrigable area, whether or not water is used

- Charges are not uniform across States; they are levied on area, not on the cost of supply or the volume used, and were fixed long ago at low levels despite rising crop prices.
- **NCAER (Naegamvala, 1972):** basing rates on project cost or crop value is not justified; base them on the **additional net benefit** due to irrigation — net benefit after irrigation minus net benefit before.

Irrigation Commission (1972) guidelines

1. Levy rates on a crop basis (except for tubewells).
2. Relate rates to gross income, not project cost — 5–12% of gross income (the upper limit for cash crops).
3. Keep rates within the irrigators' paying capacity.
4. Keep disparities between similar regions to a minimum.
5. Classify supplies into A, B and C categories by quantity and timeliness.
6. Overall, rates should not burden general revenues.

Special cases

- **Lift from canals:** lifting adds cost, so lower rates (usually half) are charged; the 1972 Commission recommended this.
- **Tubewell water:** gives a steady supply at the best crop stages; its cost depends on power, maintenance and volume pumped, so charge by volume — measured with V-notches (Uttar Pradesh, Madhya Pradesh) or from electricity consumption (Haryana, Punjab). The 1972 Commission recommended charging by the quantity supplied.
- **Conveyance-loss disparity:** if water is measured at the tubewell, fields near the well effectively pay less and distant fields more (they lose water to seepage); lined channels or underground pipelines remove this.
- **Promotional rates:** charge lower rates where irrigation is new and raise them gradually; the 1972 Commission supported this but discouraged concessions beyond two years because of revenue loss.

Sources: ICAR eCourse 'Minor Irrigation and Command Area Development' — Dr K. Shanmuga Sundaram (TNAU), reviewed by Dr P. Balakrishnan (UAS Karnataka). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The Irrigation Commission (1972) recommended water rates of

(a) 1–2% of gross income
(b) 5–12% of gross income
(c) 20–25% of net income
(d) equal to O&M cost

Q2. A charge per hectare of irrigable area, whether or not water is used, is the

(a) water rate
(b) betterment levy
(c) irrigation cess
(d) land revenue

Q3. The Government's share in the rise in land value due to irrigation is the

(a) betterment levy
(b) cess
(c) water rate
(d) tax

Q4. Lift irrigation from canals is usually charged at about

(a) double rate
(b) the same rate
(c) half rate
(d) no charge

Q5. Irrigation in India is

(a) a Union subject
(b) a State subject
(c) a concurrent subject
(d) a local-body subject only

Answers

Q1 — (b) 5–12% of gross income
Q2 — (c) irrigation cess
Q3 — (a) betterment levy
Q4 — (c) half rate
Q5 — (b) a State subject